

- [Start](#)
- [Elementary Einstein](#)
 - [Special relativity](#)
 - [General relativity](#)
 - [Gravitational waves](#)
 - [Black Holes & Co.](#)
 - [Cosmology](#)
 - [Relativity and the Quantum](#)
- [Spotlights on relativity](#)
 - [Special relativity](#)
 - [General relativity](#)
 - [Gravitational waves](#)
 - [Black Holes & Co.](#)
 - [Cosmology](#)
 - [Relativity and the Quantum](#)
 - [Complete Spotlights](#)
- [About Einstein Online](#)
 - [The concept of EO](#)
 - [About Albert Einstein](#)
 - [About Us](#)
 - [Contributing Institutions](#)
 - [FAQ](#)
 - [Copyright policy](#)
 - [Contact](#)
- [Essentials](#)
 - [Links](#)
 - [Glossary](#)
 - [Further reading](#)
- 
- 



Relativity and the Quantum / Elementary Tour: Conclusion

The combination of relativity and quantum theory has led to some of the greatest triumphs of theoretical physics, but also to its most persistent unsolved questions.

The marriage of special relativity and quantum concepts leads to so-called [relativistic quantum field theories](#) – the basis of modern particle physics – and to a variety of predictions that have passed rigorous experimental tests.

On the other hand, the question of how a union of general relativity and [quantum theory](#) should look like – a theory of [quantum gravity](#) – has not yet found a complete and convincing answer. And as long as this question remains unanswered, so do a number of the most fundamental questions physics can ask: What happens inside a black hole? What happened at the very beginning of the universe as we know it?

Physicists have found promising candidates for a theory of quantum gravity, from loop quantum gravity with its discrete spacetime structure to string theory with its promise to describe not only gravity, but all forces and matter in the same unified quantum way. However, so far none of these candidates give a complete answer to the question of quantum gravity.

This ends our brief tour of the relativistic world.

Readers who have come this far can either return to the [homepage](#) of *Einstein Online* or jump directly to the category [Relativity and the quantum](#) of our [Spotlights on relativity](#), where more intriguing facts and explanations concerning relativistic physics can be found.

[Previous part: Superstrings and universal harmony](#)
[Tour index](#)

Tag Cloud

[black holes](#) [Black Holes & Co.](#) [Cosmology](#) [Detecting](#)
[gravitational waves](#) [Einstein](#) [Energy](#) [General](#)
[relativity](#) [General theory of relativity](#)
[Geometry](#) [Gravitation](#) [Gravitational wave sources](#)
[Gravitational waves](#) [How to track compact objects](#) [Light](#)
[in general relativity](#) [Mass](#) [Matter](#) [Motion](#) [Relativity and the](#)
[Quantum](#) [Spacetime](#) [Special relativity](#) [Special relativity and](#)
[time](#) [String theory](#) [The basics of general relativity](#) [Time](#)

What is Einstein Online?

Einstein Online is a website with ...

What is Einstein Online?

Einstein Online is a web portal with comprehensible information on Einstein's theories of relativity and their most exciting applications from the smallest particles to cosmology.

Einstein Online is provided by the

[Max Planck Institute for Gravitational Physics](#)
([Albert-Einstein-Institut](#))

[About Einstein Online](#)

Includes information on our [authors](#) and [contributing Institutions](#), and a [brief history](#) of the website.

[Glossary](#)

More than 400 entries from "absolute zero" to "XMM Newton" - whenever you see [this type of link](#) on an Einstein Online page, it'll take you to an entry in our relativistic dictionary.

[a](#) [b](#) [c](#) [d](#) [e](#) [f](#) [g](#) [h](#) [i](#) [j](#) [k](#) [l](#) [m](#) [n](#) [o](#) [p](#) [q](#) [r](#) [s](#) [t](#) [u](#) [v](#) [w](#) [x](#) [y](#) [z](#)

[Links & Literatur](#)

Our recommendations for [books](#) and [websites](#) on relativity and its history.

© Max Planck Institute for Gravitational Physics, Potsdam

[Legal notice](#) | [Privacy policy](#)

